Patterns Everywhere

Mathematical and computational modeling in the Czak Research Group

Jason Czak

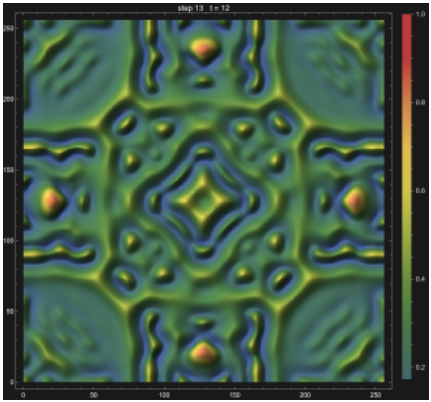
James Madison University
czakje@jmu.edu

Patterns are everywhere

Patterns appear in many physical and biological systems:

- stripes, spots, and branching structures,
- waves and chemical concentrations,
- dendrites and growing interfaces,
- disease spread and population structure,
- spatial patterns in forests.

How can local interactions produce organized structure across an entire system?



What our group investigates

We mathematically and computationally model physical systems.

The particular system depends on what question is scientifically interesting:

- How does a pattern form?
- When is the pattern stable or predictable?
- How does a local change affect the larger system?
- What can simulation or data reveal about the underlying physics?

Shared approach

Build a model, compute its behavior, test the method carefully, and connect the result to a physical question.

A starting model: the Gray–Scott reaction

The Gray–Scott model describes two concentrations, U and V, that diffuse and react:

$$U + 2V \longrightarrow 3V,$$
$$V \longrightarrow Z.$$

Material is supplied and removed while the two concentrations interact locally.

Why it is interesting

A small reaction model can produce spots, waves, periodic structures, and irregular behavior.

P. Gray and S. K. Scott, *Chemical Oscillations and Instabilities: Non-linear Chemical Kinetics* (1990).

The Gray–Scott equations

$$\frac{\partial u}{\partial t} = D_u \nabla^2 u - uv^2 + A(1-u),$$

$$\frac{\partial v}{\partial t} = D_v \nabla^2 v + uv^2 - (A+B)v.$$

Diffusion	**Reaction**	**Flow**
material spreads through space	U and V interact nonlinearly	material enters and leaves the system

From equations to a simulation

We divide space into lattice points and approximate the spatial derivative:

$$\nabla^2 U_i^t \approx \frac{U_{i+1}^t + U_{i-1}^t - 2U_i^t}{h_x^2}.$$

A simple Euler update has the form

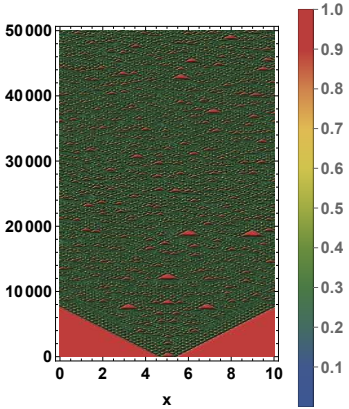
$$U_i^{t+1} = U_i^t + h_t\,(\text{diffusion} + \text{reaction} + \text{flow})\,.$$

1. Calculate the change at each lattice point.
2. Move forward by one small time step.
3. Repeat to produce the evolving pattern.

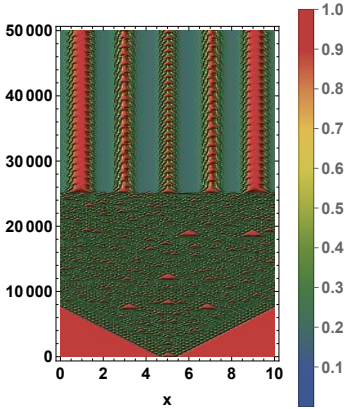
Important lesson

A simulation is a numerical experiment. The time step, grid spacing, and numerical method must be checked before trusting the result.

Parameter-guided pattern manipulation



Unperturbed evolution



A local change is applied

Gray–Scott as a training model

Gray–Scott is a useful place for a new group member to begin.

Students can use it to learn how to:

- read a mathematical model,
- turn equations into working code,
- visualize and measure patterns,
- test numerical reliability,
- develop a focused research question.

The purpose of the training

The goal is to build the technical foundation needed to contribute reliably to a difficult research problem.

Where that foundation can lead

- Electric-field effects
- Numerical stability and sensitivity
- Spatially varying model parameters
- Prediction of evolving patterns

Training $\longrightarrow$ stronger skills $\longrightarrow$ guided contribution

Other group directions

- Disease and epidemic models
- Dendrite growth with U.S. Naval Academy collaborators
- Tree and forest patterns from ecological data

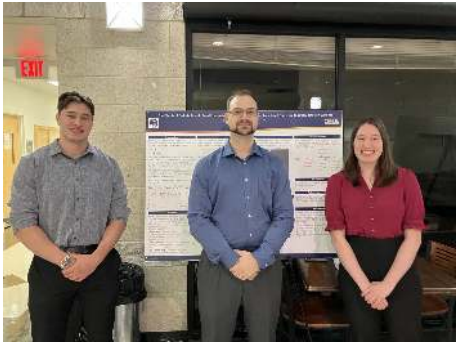
Different systems, connected by mathematical modeling and computation

What it is like to start in the group

You begin with a starter research paper and a manageable first task.

- Ask questions about the physics, mathematics, paper, or code.
- Receive substantial help while building the required skills.
- Learn through coding, debugging, testing, and discussion.

No generative AI or LLM use during initial training.



Why no LLM during initial training?

The early goal is to learn how the equations, code, and physics connect.

- Understand what each part of the program does.
- Recognize numerical and physical mistakes.
- Explain why a result should or should not be trusted.

Support remains central

The restriction is on outsourcing the reasoning, not on asking for help.

How the work develops

1. Read a starter paper and learn the model.
2. Reproduce and explain an established result.
3. Build and test the computational tools.
4. Contribute to a larger problem with continued guidance.

The aim is steady technical growth and a useful contribution by graduation.

Interested in joining the group?

You do not need previous experience with all of the mathematics or computational tools.

A good starting point is:

- curiosity about how physical systems behave,
- willingness to ask questions,
- patience with coding and debugging,
- readiness to build skills step by step.

Contact: `czakje@jmu.edu`



Questions?

What kind of physical system or computational problem would you want to explore?

Complete Euler update

$$U_i^{t+1} = U_i^t + h_t\left(D_u \frac{U_{i+1}^t + U_{i-1}^t - 2U_i^t}{h_x^2} - U_i^t (V_i^t)^2 + A(1 - U_i^t)\right),$$

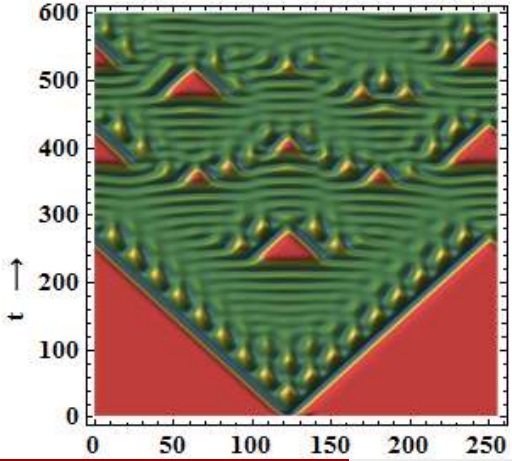
$$V_i^{t+1} = V_i^t + h_t\left(D_v \frac{V_{i+1}^t + V_{i-1}^t - 2V_i^t}{h_x^2} + U_i^t (V_i^t)^2 - (A + B)V_i^t\right).$$

Numerical question

How small must h_t and h_x be before the interpretation of the pattern stops changing?

Numerical-method comparison

Euler



Runge–Kutta

